OCEANRESCUE
PROJECT MASTER DOCUMENT

Every Net Found. Every Life Saved.

Official Technical Title

AI-Driven Geospatial Intelligence for Detection, Drift Forecasting, and Retrieval Optimization of ALDFG (Abandoned, Lost, and Discarded Fishing Gear)

Third Year Mini Project

B.E. Computer Engineering

Academic Project Document

Project positioning: An AI-driven marine conservation and decision-support approach connecting detection, geospatial intelligence, drift forecasting, risk prioritization and recovery-route planning.

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1. PROJECT IDENTITY

Project Name	OceanRescue
Tagline	Every Net Found. Every Life Saved.
Official Technical Title	AI-Driven Geospatial Intelligence for Detection, Drift Forecasting, and Retrieval Optimization of ALDFG (Abandoned, Lost, and Discarded Fishing Gear)
Domain	Artificial Intelligence • Computer Vision • GeoAI • GIS • Marine/Ocean Intelligence
Project Type	Third Year B.E. Computer Engineering Mini Project
One-line Description	An intelligent decision-support approach for detecting potential ALDFG, mapping it, forecasting probable movement, prioritizing recovery and recommending efficient retrieval routes.

2. PROJECT OVERVIEW

Abandoned, Lost, and Discarded Fishing Gear (ALDFG) is a persistent marine-debris problem because nets, lines, traps and related equipment can remain in the ocean after being lost or discarded. Such gear may continue capturing marine organisms without human control, damage habitats and create navigation or fishing hazards.

OceanRescue is positioned as a predictive, geospatially driven decision-support approach rather than a standalone image-classification project. Its intended workflow connects AI-based detection with geographic localization, environmental data integration, drift forecasting, risk prioritization and recovery-route planning.

Core idea: Detect → Locate → Predict → Prioritize → Optimize → Recover.

The project is designed for prototype-scale implementation. Proposed functionality must be validated with suitable research datasets and defined test scenarios before claims about real-world performance are made.

3. DETAILED PROBLEM STATEMENT

• Fishing gear can be lost because of storms, strong currents, equipment failure, gear conflicts, difficult retrieval conditions or accidental loss during fishing operations.

• Damaged or unwanted gear may be improperly discarded, adding persistent fishing equipment to marine debris.

• Lost gear can trap fish, turtles, seabirds, dolphins, whales and other marine organisms, contributing to ghost fishing.

• ALDFG can damage coral reefs, seagrass, seabed habitats and other marine ecosystems.

• Drifting gear can create navigation hazards by becoming entangled with vessels or equipment.

• Currents, tides, wind and waves can move gear away from its last-known location.

• Manual recovery using boats, divers, visual inspection and sonar can be costly, time-consuming and geographically limited.

• Knowing where gear is today is not enough; recovery teams also need to estimate where it may move and which targets should be addressed first.

Core problem: *How can AI, geospatial information and ocean/environmental data be combined to detect potential ALDFG, estimate its location, forecast probable movement, assess recovery priority and support efficient retrieval planning?*

4. COMMUNITY / STAKEHOLDERS

Stakeholder	Role / Need
Fishing Communities	Report last-known gear locations; provide local knowledge; benefit from reduced gear-loss hazards and cleaner fishing grounds.
Fisheries Departments / Marine Authorities	Monitor hotspots, coordinate recovery, prioritize resources and support responsible gear management.
Marine Conservation Organizations / NGOs	Use detections and drift forecasts to target cleanup operations and track recovery outcomes.
Coast Guard / Maritime Authorities	Use ALDFG information as a navigation and maritime-monitoring input and coordinate with recovery teams.
Recovery / Cleanup Teams	Use locations, predicted movement, risk levels and recommended routes to plan physical retrieval.
Coastal Communities	Benefit from healthier marine ecosystems, safer waters and protection of fishing, tourism and livelihood interests.
Marine Researchers / Ocean Scientists	Use historical detection and drift information for research, validation and improved forecasting.
Government & Environmental Policy Makers	Use aggregated hotspot and recovery information to support monitoring, planning and policy development.

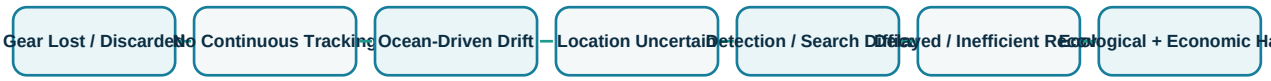
Stakeholder flow: Fishermen → Report / last-known location → OceanRescue → Authorities / NGOs → Recovery Teams / Vessels → Cleaner and safer marine environment.

5. ROOT CAUSE ANALYSIS

Cause	Explanation
Accidental Loss of Fishing Gear	Storms, currents, high waves, equipment failure and operational conditions can separate gear from vessels.
Improper Disposal	Damaged or unwanted gear may be abandoned or discarded.
Difficult Marine Conditions	Currents, tides, wind and waves transport gear away from its original location.
Lack of Real-Time Gear Tracking	Vessel tracking does not normally provide continuous tracking of individual lost gear.
Difficulty Detecting Submerged Gear	Depth, turbidity, darkness, seabed conditions and vegetation can hide gear.
Dependence on Manual Search	Boats, divers, sonar and visual inspection require time, fuel and expertise.
Lack of Predictive Monitoring	A current detection does not automatically provide a probable future location.
Fragmented Data	Imagery, sonar, GPS, currents, weather and fishing information may be separate.
Inefficient Retrieval Planning	Multiple targets require prioritization and route planning beyond simple distance.
Environmental Consequences	Persistent gear can cause ghost fishing, habitat damage, pollution and economic/navigation impacts.

Root cause: *The absence of a continuous, intelligent monitoring workflow connecting detection with location, probable future movement, risk assessment and recovery planning.*

Root-cause chain



6. EXISTING SOLUTIONS

Approach	What it provides	Key limitation
Manual Boat Surveys	Visual search and physical recovery.	Large-area searches are costly and slow.
Diver Surveys	Direct underwater inspection.	Depth, visibility, current and safety constraints.
Side-Scan Sonar	Underwater seabed/object signatures.	Expert interpretation and follow-up verification may be needed.
Underwater Cameras / ROVs	Visual inspection at depth.	Turbidity, darkness and water conditions affect visibility.
Satellite / Remote Sensing	Large-area environmental and surface monitoring.	Small or submerged gear may be difficult to detect reliably.
GPS / AIS / VMS	Vessel location and fishing-activity context.	Generally tracks vessels, not individual lost gear.
Fishermen Reporting	Last-known location and gear information.	Location becomes less reliable as gear drifts.
Ocean Drift Models	Probable movement from environmental conditions.	Accuracy depends on data and representation of gear behavior.
Organized Retrieval Programs	Targeted cleanup and physical removal.	Often campaign-based rather than continuously automated.

Key observation: Existing technologies solve important individual stages, but they are not necessarily integrated into one workflow from detection through forecasting, prioritization and retrieval planning.

7. EXISTING-SOLUTION LIMITATIONS

• Manual detection is time-consuming.

• Boats, fuel, divers, sonar and specialized recovery teams can create high operational cost.

• Manual surveys have limited geographic coverage.

• Submerged, buried or entangled gear is difficult to detect visually.

• Satellite imagery may not reliably reveal small, submerged or partially hidden gear.

• Sonar can identify suspicious structures but may require expert interpretation.

• Underwater camera performance depends on turbidity, light, depth and water quality.

• Individual lost gear is generally not continuously tracked like a vessel.

• Drift prediction is uncertain because marine conditions and gear behavior vary.

• Data from imagery, sonar, GPS, weather and ocean sources can be fragmented.

• Many workflows still depend on human observation and interpretation.

• Existing approaches may not automatically prioritize targets using ecological and operational risk.

• Recovery planning may not jointly consider multiple targets, vessel position, priorities and predicted locations.

• Real-time decision support may be limited when information is spread across separate tools.

Main limitation: *There is a gap in integrating automated detection, geospatial localization, spatiotemporal drift forecasting, risk prioritization and retrieval optimization into one decision-support workflow.*

8. PROPOSED SOLUTION — OCEANRESCUE

1. Multi-Source Data Collection

Use suitable underwater/visual imagery, satellite or remote-sensing information, sonar where available, fishermen reports, GPS and environmental/ocean data.

2. AI-Based Detection

Use a YOLO-family object detector selected and trained according to available labelled data. Output may include gear class, bounding box and confidence.

3. Geospatial Localization

Associate detections with latitude, longitude, time, source and confidence where the underlying data permits.

4. Ocean Data Integration

Combine detection information with currents, wind, waves, tides, weather and bathymetry.

5. Drift Forecasting

Estimate probable future positions at selected horizons such as +6, +12 and +24 hours, represented as a path/corridor with uncertainty.

6. Risk & Priority

Rank targets using defined ecological and operational factors.

7. Retrieval Optimization

Recommend an efficient recovery sequence/route based on vessel position, target priority, predicted locations and constraints.

8. GIS Decision Support

Present detections, drift paths, risk, vessel location, route and recovery status in one interactive interface.

9. Recovery Feedback

Record recovery outcomes and compare predicted and actual information for evaluation and future improvement.

Boundary: OceanRescue does not claim autonomous physical retrieval. Retrieval optimization means recommending and prioritizing recovery actions/routes for qualified recovery teams or vessels.

9. TECHNICAL MODULES

Module	Purpose	Main technologies / outputs
ALDFG Detection	Identify nets, ropes/lines, traps/pots and other dataset-supported gear.	YOLO-family, OpenCV, PyTorch; class, box, confidence.
Geospatial Localization	Attach detections to geographic coordinates and time.	Python, GeoPandas, Shapely.
Ocean Data Integration	Combine environmental conditions with observations.	Currents, wind, waves, tides, weather, bathymetry.
Drift Forecasting	Estimate probable future locations and trajectories.	ML/deep-learning as justified by data; uncertain corridor.
Risk Assessment	Assign recovery priority using ecological/operational factors.	Defined scoring/rule/ML approach; Low–Critical.
Retrieval Optimization	Recommend recovery sequence and route.	A*/Dijkstra/VRP-style optimization as appropriate.
GIS Dashboard	Visualize detections, drift, risk, vessel and route.	React, Leaflet/MapLibre.
Backend / API / Database	Connect models, spatial data and dashboard.	FastAPI, PostgreSQL + PostGIS.

10. END-TO-END WORKFLOW

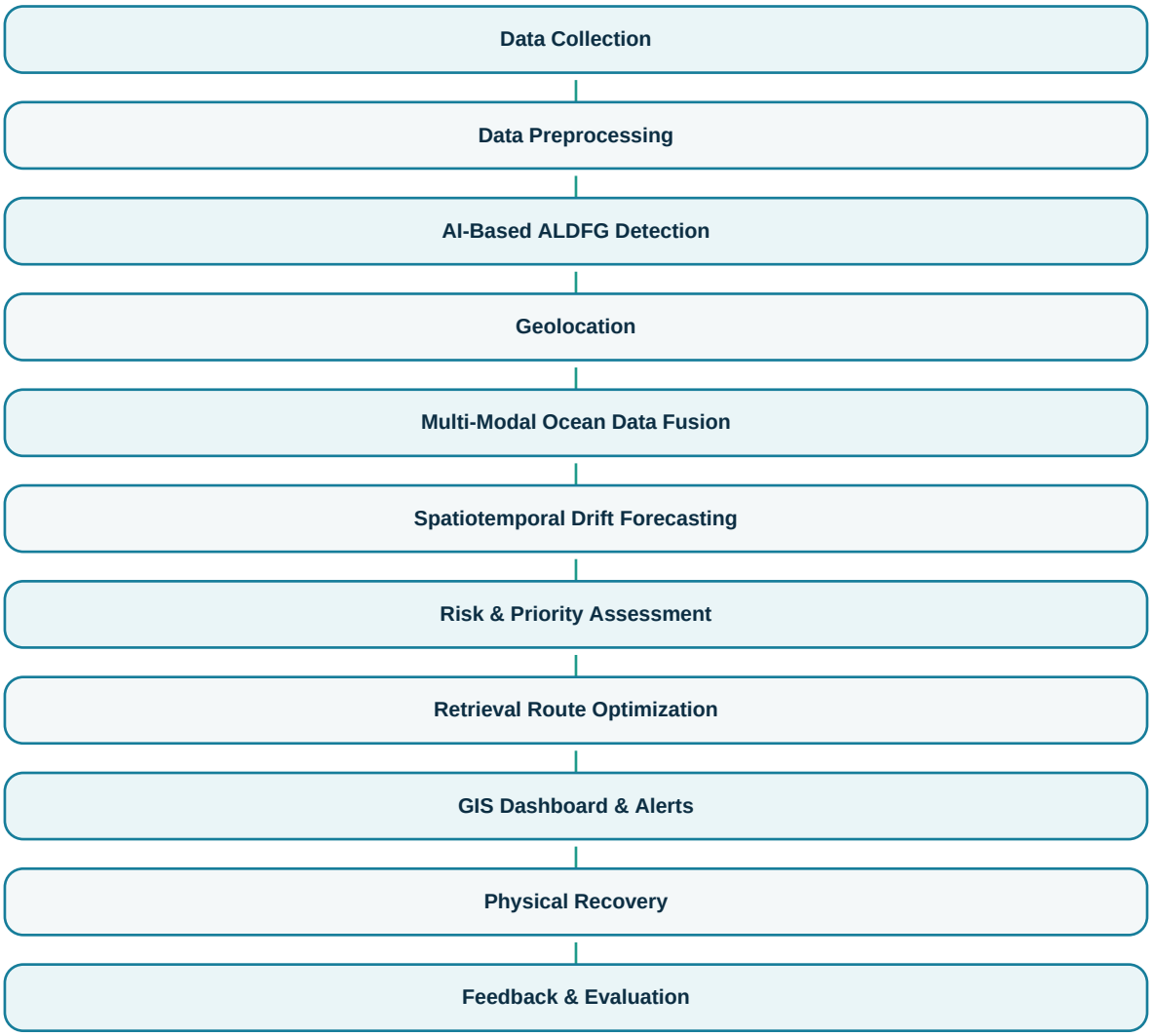


Figure 1. OceanRescue end-to-end workflow.

Simple example: Lost net → AI detects potential fishing net → location associated → ocean data analysed → movement forecast → high-priority target → recovery route recommended → team attempts retrieval → dashboard updated as recovered.

11. AI / TECHNOLOGY STACK

Component	Technology	Purpose
Programming	Python	AI/ML, data, geospatial processing and forecasting.
ALDFG Detection	YOLO-family model	Visual detection of supported gear classes.
Computer Vision	OpenCV	Image processing.
Deep Learning	PyTorch	Training and evaluation.
ML Baselines	Scikit-learn	Classical ML baselines/evaluation.
GeoAI	GeoPandas, Rasterio, Shapely	Vector/raster/spatial processing.
GIS	QGIS	Spatial analysis and layer preparation.
Drift Forecasting	ML; LSTM/GRU if justified	Spatiotemporal prediction.
Ocean Data	Current, wind, wave, tide, weather datasets/APIs	Environmental inputs.
Route Optimization	A*/Dijkstra and/or VRP	Route/sequence recommendation.
Database	PostgreSQL + PostGIS	Spatial storage and queries.
Backend	FastAPI	APIs.
Frontend	React.js	Dashboard.
Map	Leaflet / MapLibre	Geographic visualization.
Optional IoT	ESP32 + GPS	Vessel-position demonstration.

Technology principle: The project does not require Blockchain, Generative AI, AR/VR or unrelated technologies unless a later requirement gives them a genuine purpose.

12. DATA REQUIREMENTS

Data Type	Use	Status / Note
ALDFG image data	Detection training/testing.	Required; dataset must be identified and verified.
Underwater imagery/video	Visual detection.	Useful where quality/labels permit.
Sonar observations	Submerged-object context.	Optional; research datasets may be used.
Satellite / remote sensing	Large-area context.	Use where appropriate.
Fishermen reports	Last-known location/context.	Useful; location may become uncertain.
GPS / vessel position	Localization and route planning.	Required for route scenarios.
Ocean currents	Drift forecasting.	Key forecasting input.
Wind / waves / tides	Environmental forcing.	As available/appropriate.
Weather	Operational/environmental context.	Useful input.
Bathymetry / depth	Spatial/environmental context.	Useful input.
Sensitive marine areas	Risk prioritization.	Required if used in scoring.
Fishing / shipping activity	Operational risk context.	Only where reliable data exists.

13. DATA FLOW

OceanRescue treats data as a sequence of linked but separately validated inputs. Each stage should preserve source metadata, timestamps and uncertainty where possible.

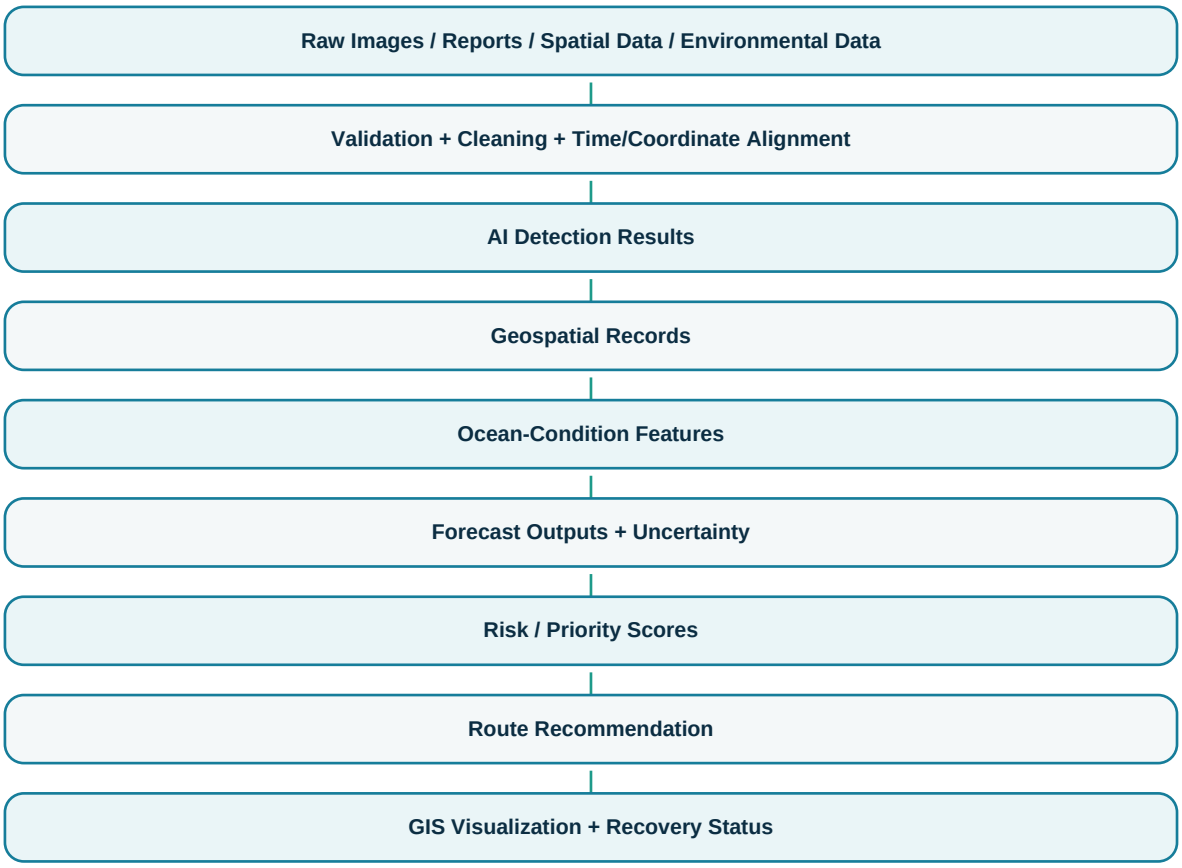


Figure 2. Conceptual data flow and transformation stages.

A database record should retain enough provenance to answer what was detected, when, from which source, at what location, with what confidence, and what prediction or recovery status is currently associated.

14. SYSTEM/PROJECT ARCHITECTURE

The following is an **architecture diagram** of the project. The project itself is not being described as a “System” or “Framework”; the term is used only as a technical diagram heading.

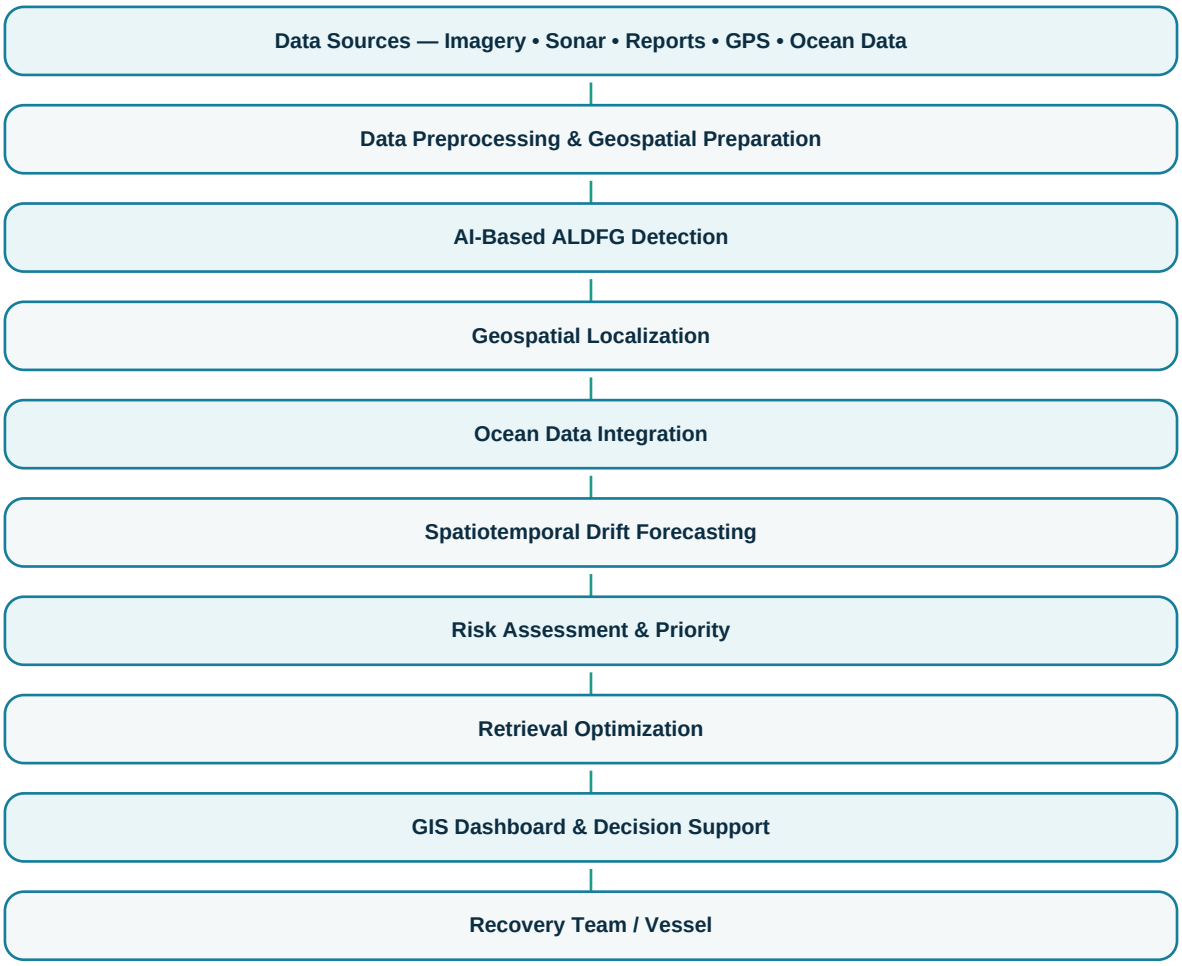


Figure 3. Overall OceanRescue architecture.

Feedback loop: Recovery outcomes and actual observations can be stored for evaluation and potential model improvement. Automatic retraining is not assumed unless implemented and validated.

15. AI/ML PIPELINE



Figure 4. AI/ML pipeline for ALDFG visual detection.

Dataset

Identify legally usable, relevant ALDFG imagery with sufficient class coverage.

Preprocessing

Resize/normalize images and apply quality checks and appropriate augmentation.

Annotation

Create or verify bounding-box labels for supported ALDFG classes.

Training

Train a selected YOLO-family model using a defined train/validation/test strategy.

Evaluation

Report precision, recall, F1-score and mAP on a held-out test set; do not invent results.

Detection

Run inference on suitable imagery and return class, bounding box and confidence.

16. GEOSPATIAL INTELLIGENCE FLOW

Observation	Coordinates / geometry	Context
Image detection	GPS or image metadata where available	Time, source, confidence
Reported loss	Fishermen-provided coordinates	Gear type, loss time, report source
Environmental layers	Raster/vector spatial data	Currents, wind, bathymetry, sensitive areas
Target record	Point/line/polygon as appropriate	Risk, forecast and recovery status

A key design principle is to preserve location uncertainty rather than convert uncertain historical reports into falsely precise points.

17. DRIFT FORECASTING APPROACH

Goal: Estimate probable future ALDFG positions from current/last-known location, time and environmental conditions.

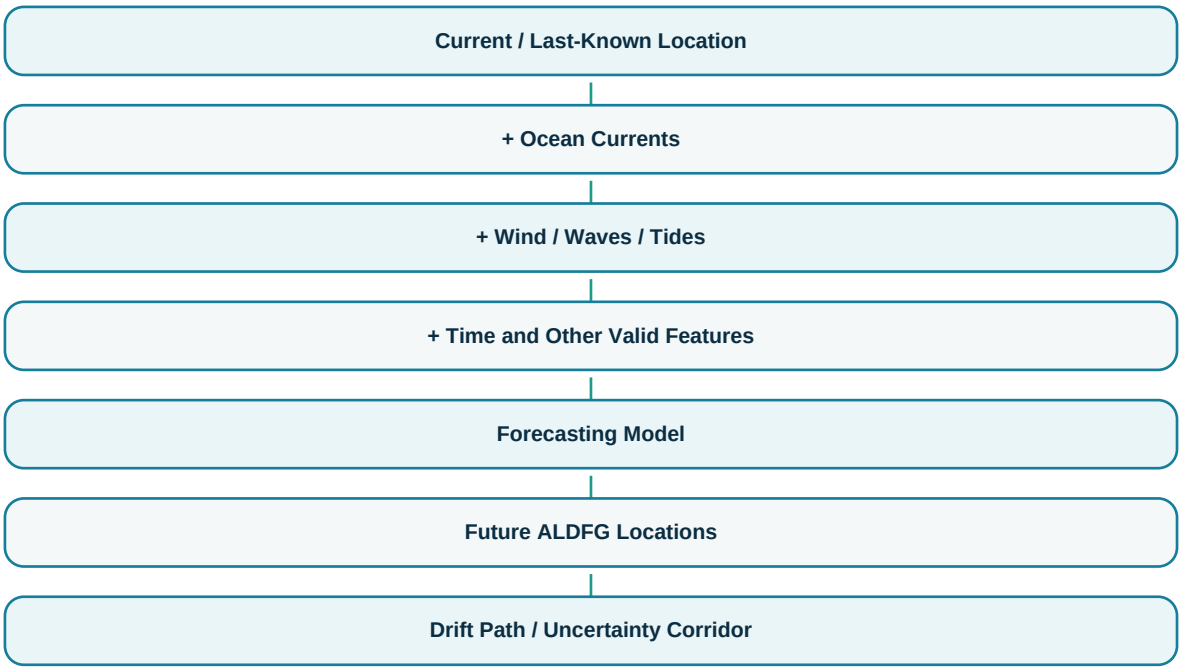


Figure 5. Conceptual drift-forecasting flow.

A practical strategy is to begin with a transparent baseline such as Random Forest or XGBoost if the data is tabular and sufficient; consider LSTM/GRU only if the dataset supports meaningful sequential learning. Model selection should be based on validation performance, not complexity.

Forecast outputs may use +6, +12 and +24 hour horizons. Because ocean movement is uncertain, results should be presented as probable locations or corridors rather than guaranteed exact trajectories.

18. RETRIEVAL OPTIMIZATION APPROACH

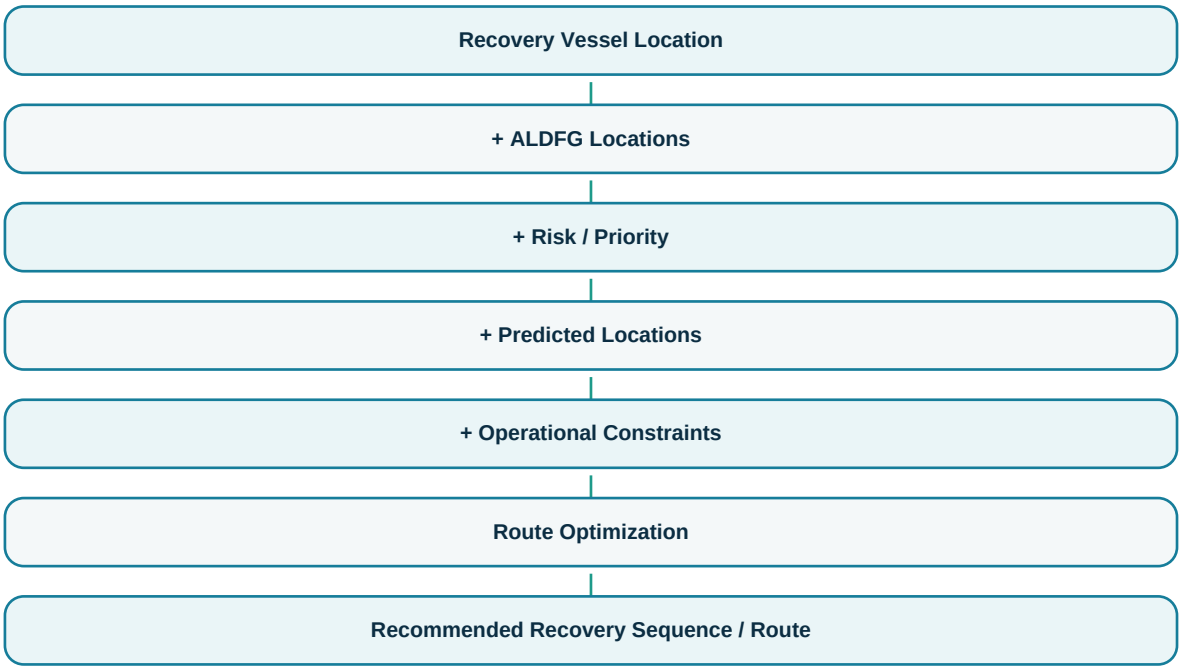


Figure 6. Conceptual retrieval optimization flow.

The objective is to recommend a practical recovery sequence while considering distance, predicted locations, priority and operational constraints. A*/Dijkstra may suit a single-route problem; a vehicle-routing formulation may suit multiple targets.

Boundary: The output is a recommendation for recovery teams/vessels; the project does not claim autonomous physical retrieval.

19. GIS DASHBOARD CONCEPT

Map Element	Conceptual Representation
ALDFG detection points	Markers showing gear type, confidence and timestamp.
Current / last-known location	Primary marker with source and uncertainty where applicable.
Predicted drift paths	Lines/corridors showing forecast positions and uncertainty.
Risk zones	Overlays representing defined ecological/operational risk.
Vessel location	Live or simulated recovery-vessel GPS marker.
Recommended route	Highlighted route and ordered recovery targets.
Recovery status	Active / prioritized / recovered states.
Alerts and statistics	Priority notifications and summary counts.

Figure 7. GIS dashboard concept: detections + drift + risk + vessel + route + recovery status.

20. OBJECTIVES

• Develop a Computer Vision approach to detect identifiable ALDFG classes from suitable data.

• Associate detections with latitude, longitude, timestamp, gear type and confidence where supported.

• Integrate detections with currents, wind, waves, tides, weather and bathymetry.

• Develop and evaluate a spatiotemporal approach for probable future locations and drift paths.

• Assess environmental and operational risk and prioritize recovery targets.

• Generate efficient recovery sequences/routes using vessel position, predicted locations and priority.

• Develop a GIS-based decision-support dashboard.

• Evaluate detection, forecasting, route efficiency and end-to-end workflow using defined scenarios.

• Support faster and more targeted recovery planning.

In one line: *Detect → Locate → Predict → Prioritize → Optimize → Recover.*

21. INNOVATION ASPECT

Innovation	Contribution
GeoAI-Based Detection	Combines object detection with geographic information.
Multi-Modal Ocean Intelligence	Connects imagery, sonar where available, GPS and environmental data.
Spatiotemporal Drift Forecasting	Extends the question from where gear is now to where it may move over time.
Risk-Based Prioritization	Ranks targets using ecological and operational factors.
AI-Assisted Retrieval Optimization	Uses predicted locations and vessel position to recommend recovery sequence.
Detection-to-Recovery Pipeline	Connects detection, localization, forecasting, risk, route planning and status.
Dynamic GIS Decision Support	Shows target, risk, route and recovery information on a map.
Live/Updated Data Integration	Can incorporate updated environmental conditions and optional ESP32 + GPS.
Predictive Rather Than Reactive	Moves from search-after-loss toward forecasting and targeted planning.

Core innovation: *Integration of GeoAI, multi-modal ocean data, spatiotemporal drift forecasting, risk-based prioritization and retrieval-route optimization.*

22. SCOPE

• Detect nets, ropes/lines, traps/pots and other dataset-supported gear.

• Prototype on selected coastal/fishing regions rather than assuming global-ocean coverage.

• Integrate underwater imagery, sonar where available, satellite/remote sensing, reports, GPS and environmental data.

• Provide GIS monitoring of detections and potential hotspots.

• Forecast probable movement and display uncertainty corridors.

• Assess risk near sensitive habitats, fishing areas, shipping/navigation routes and coastal regions where reliable layers exist.

• Recommend recovery priorities and routes for recovery teams/vessels.

• Provide an interactive GIS dashboard.

• Optionally demonstrate vessel-position updates with ESP32 + GPS.

• Leave room for future expansion to additional gear, regions, sensors, AUVs and advanced models.

23. LIMITATIONS

- Limited high-quality labelled ALDFG underwater/sonar datasets.
- Underwater turbidity, darkness, depth and particles can reduce detection quality.
- Satellite imagery may not reliably detect small, submerged or deep gear.
- Sonar signatures can be ambiguous and may require expert interpretation.
- Exact geolocation may be uncertain when observations lack precise GPS.
- ALDFG drift depends on changing currents, wind, waves, tides, depth and gear characteristics.
- Forecasting quality depends on environmental-data accuracy, resolution and update frequency.
- Models may generalize differently across regions and conditions.
- The prototype may rely on historical or periodically updated data rather than continuous real-time feeds.
- Physical retrieval depends on vessels, equipment, permissions and trained personnel.
- Recommended routes may change because of weather, traffic, restrictions or updated forecasts.
- Optional hardware/AUV work introduces waterproofing, communication, battery and sensing constraints.
- Deep-learning and geospatial workloads may require GPU/compute resources.
- Real-world field validation can be difficult because of access, safety and marine logistics.

Main limitation: Marine conditions and limited ALDFG data create uncertainty in detection and drift prediction. The project should communicate probabilistic predictions and decision support rather than claim 100% detection or exact future locations.

24. RESOURCES REQUIRED

Category	Resources
Hardware	Laptop/PC; GPU access recommended for deep-learning training.
Optional Hardware	ESP32 + GPS; underwater camera/ROV; sonar; basic electronics.
Software	Python, PyTorch, OpenCV, Scikit-learn, YOLO-family, GeoPandas, Rasterio, Shapely, QGIS, FastAPI, React, Leaflet/MapLibre, PostgreSQL + PostGIS.
Data	ALDFG imagery; underwater images/video; optional sonar; satellite/remote sensing; GPS/history; currents; wind/waves; tides/weather; bathymetry; sensitive-area layers; fishing/shipping activity where available.
AI/ML	Labelled train/validation/test data, compute, evaluation metrics, historical spatiotemporal data and baselines.
Cloud / Version Control	Cloud GPU/server hosting if required; Git and GitHub.
Human Resources	AI/ML, GIS, full-stack, marine-domain guidance, project guide/team.
Research Resources	Peer-reviewed literature, government/marine-organization datasets and reports.

Minimum prototype: Laptop/PC + Python + suitable ALDFG dataset + environmental data + YOLO-family model + PostGIS + FastAPI + React + Leaflet/MapLibre.

25. RESEARCH / LITERATURE REVIEW

Study	Publication / Journal	Key Contribution	Relevance
Richardson et al. (2019)	Building evidence around ghost gear: Global trends and analysis for sustainable solutions at scale — Marine Pollution Bulletin	Global evidence, standardized ALDFG data, retrieval and fragmented information.	Supports structured, location-aware ALDFG data and management.
Do & Armstrong (2023)	Ghost fishing gear and their effect on ecosystem services – Identification and knowledge gaps — Marine Policy	Systematic review/meta-analysis identifying ecological and socioeconomic knowledge gaps.	Supports broader data-driven ALDFG management.
Continental Shelf Research (2023)	Evaluating the use of side scan sonar for improved detection and targeted retrieval of abandoned, lost, or otherwise discarded fishing gear	SSS for local-scale detection and targeted retrieval; modelling and fisher knowledge.	Supports sonar as one component while motivating integration with forecasting.

Reference metadata and DOI details should be verified against the publisher or scholarly index before final academic submission.

26. RESEARCH GAP

Gap	Opportunity
Detection and recovery are often separate	Connect finding and recovery planning in one workflow.
Limited multi-modal integration	Combine imagery, sonar, GPS, currents, wind, tides and weather.
Need for spatiotemporal prediction	Estimate where gear is likely to be later.
Limited risk-based prioritization	Decide which target deserves attention first.
Limited retrieval optimization	Connect predicted targets with recovery-vessel position and route planning.
Limited end-to-end decision support	Connect detection → geolocation → forecasting → risk → optimization → GIS.

Research contribution: *OceanRescue aims to investigate an integrated decision-support workflow combining GeoAI-based detection, multi-modal ocean data, spatiotemporal drift forecasting, risk-based prioritization and retrieval-route optimization.*

27. EXPECTED OUTCOMES

• An evaluated Computer Vision prototype for identifiable ALDFG classes.

• Geospatial mapping of detections with available location, confidence, timestamp and status.

• Probable future positions/drift paths represented with uncertainty.

• Recovery-priority categories such as Low, Medium, High and Critical.

• A recommended recovery sequence/route for selected scenarios.

• An interactive GIS dashboard combining detections, drift, risk, vessel position, route and recovery status.

• An end-to-end demonstration: Detect → Locate → Forecast → Assess Risk → Prioritize → Optimize → Recover.

• Component-level and end-to-end evaluation without invented results.

• Optional ESP32 + GPS demonstration of vessel movement.

• A research-oriented prototype supporting future field validation.

Expected final result: A working decision-support prototype that can identify potential ALDFG, map its location, estimate probable movement, prioritize recovery targets and recommend an efficient retrieval route through an interactive GIS dashboard.

28. PERFORMANCE AND EVALUATION PLAN

Component	Metrics / Evaluation	Purpose
Detection	Precision, Recall, F1-score, mAP; error analysis.	Measure detection quality on held-out data.
Geolocation	Coordinate error where ground truth exists; metadata completeness.	Assess spatial reliability.
Drift Forecasting	Position/trajectory error; horizon-wise error; baseline comparison.	Measure forecast quality.
Risk Assessment	Sensitivity analysis; scenario consistency; expert/rule review where applicable.	Check intended prioritization behaviour.
Route Optimization	Distance/time; comparison with defined baseline; target sequence.	Assess route efficiency.
Dashboard/API	Response time, request success, workflow completion.	Assess prototype performance.
End-to-End	Successful execution across defined scenarios; documented failure cases.	Evaluate integration.

Evaluation principle: No accuracy, error or efficiency value should be reported unless measured from actual project data and experiments.

29. FINAL PROJECT POSITIONING

OceanRescue is an **AI-driven marine conservation and decision-support approach** for ALDFG management. It is deliberately broader than a ghost-net image detector.

Basic project: Underwater Image → AI → Ghost Net Detected



Figure 8. OceanRescue positioning compared with a standalone detection task.

Final positioning statement: *The project aims to develop an intelligent, predictive and geospatially driven approach for ALDFG management by integrating AI-based detection, multi-modal ocean intelligence, spatiotemporal drift forecasting and retrieval optimization. The objective is to help transform ALDFG recovery from a largely reactive search process into a more targeted, predictive and data-driven operation.*

30. FUTURE ENHANCEMENTS

- Expand gear classes when reliable labelled datasets become available.
- Extend validation to larger marine areas.
- Integrate additional underwater sensors and higher-quality sonar/imagery where feasible.
- Explore AUV-assisted data collection subject to safety and hardware feasibility.
- Develop more advanced probabilistic ocean-drift models.
- Develop long-term ALDFG hotspot prediction.
- Integrate government/NGO operational data where access permits.
- Explore automated recovery equipment as a separate future engineering challenge.
- Add model monitoring, data-quality checks and uncertainty visualization.
- Conduct controlled field validation when access, permissions, safety and equipment are available.

31. PROJECT SUMMARY

One-line summary

OceanRescue uses AI and geospatial/ocean intelligence to detect potential ALDFG, forecast probable drift, prioritize recovery targets and recommend efficient retrieval routes.

Three-line summary

OceanRescue connects Computer Vision with GeoAI and multi-modal ocean data. It estimates where detected gear may move, assesses risk and prioritizes recovery targets. A GIS dashboard supports recovery teams with mapped targets, forecasts and route recommendations.

Academic paragraph

OceanRescue is a Third Year B.E. Computer Engineering mini project at the intersection of artificial intelligence, geospatial computing and marine environmental data. The proposed approach addresses the operational challenge that ALDFG can be difficult to detect, may drift after loss, and can require costly search and recovery operations. By integrating visual detection with geospatial localization, environmental data fusion, spatiotemporal forecasting, risk prioritization and retrieval optimization, the project investigates how disparate information sources can be transformed into actionable decision support. The prototype is intended to be evaluated using defined datasets and scenarios, with uncertainty explicitly represented and without claiming autonomous physical recovery or unverified real-world performance.

30-second explanation

“Our project is OceanRescue. It addresses abandoned, lost and discarded fishing gear, or ghost gear. Instead of only detecting a net, we connect detection with its location, ocean conditions and probable future movement. We then assess which gear is more urgent and recommend an efficient recovery route for a vessel or recovery team. The final result is an interactive GIS decision-support dashboard.”

Viva explanation

Problem: Lost fishing gear can continue harming marine life and is difficult to find after drifting. **Solution:** OceanRescue combines AI-based visual detection, geospatial processing, ocean-data integration, drift forecasting, risk prioritization and retrieval optimization. **Output:** A GIS dashboard showing detected gear, probable drift, risk and a recommended recovery route. **Innovation:** The value is in connecting these stages into one predictive workflow rather than building only a detection model. **Boundary:** The project supports human recovery teams; it does not claim autonomous physical retrieval.

32. REFERENCES

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[5] QGIS Documentation. Official documentation: https://docs.qgis.org/

[6] PostGIS Documentation. Official documentation: https://postgis.net/docs/

[7] FastAPI Documentation. Official documentation: https://fastapi.tiangolo.com/

[8] GeoPandas Documentation. Official documentation: https://geopandas.org/

Reference verification note: Publisher metadata and DOI details should be verified directly before final submission and formatted according to the department's required citation style.

DOCUMENT STATUS

Planning / Master Reference — Pre-Development

This document consolidates the project's current concept, scope, architecture, proposed technologies, research motivation and evaluation plan. Proposed functionality is not represented as completed implementation. Experimental results, model accuracy, dataset performance and field validation must be added only after actual development and testing.

OCEANRESCUE

Every Net Found. Every Life Saved.